

Kyaw Soe Lwin

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EDUCATION

University of California San Diego

B.S. Data Science

Jun 2028

GPA: 4.0

Coursework: Machine Learning, Deep Learning, Computer Vision, Statistical Natural Language Processing

EXPERIENCE

Data Science Alliance

Data Science Volunteer

San Diego, CA

Sep 2026 – Present

- Designed and executed an end-to-end data auditing and validation pipeline, cross-referencing multi-year counts against source reports to ensure high data integrity (97.5%+ fidelity).
- Standardized schemas and built geospatial crosswalks across 380+ downtown blocks and neighborhood boundaries to enable spatial panel modeling.
- Engineer and preprocess spatial datasets across 50K+ geo-records for time-series decomposition, Getis-Ord Gi* hotspot detection, and predictive modeling.

PROJECTS

Enterprise Corrective Multimodal RAGOps Platform (GCP)

2026

- Architected an enterprise multimodal CRAG platform on GCP using Document AI Layout Parser and Vertex AI 1408-d embeddings to index text, Markdown tables, and cropped chart figures.
- Implemented a cyclic LangGraph state machine with dynamic relevance grading and grounded Gemini 2.5 Flash synthesis; evaluated via Ragas CI/CD (0.7519 Faithfulness) on Cloud Run.

LoRA TinyLlama-1.1B Instruction Fine-Tuning

2026

- Fine-tuned TinyLlama-1.1B with LoRA, adapting only the attention projection layers (q/k/v/o) to cut trainable parameters by over 95% while keeping the base model frozen.
- Used 4-bit quantization to fit the entire training run on a free Colab T4 GPU, then diagnosed and fixed a KV-cache/gradient-checkpointing conflict that was causing garbled output at inference.

Telco Customer Churn Pipeline & Profit Thresholding

2026

- Engineered a leakage-audited preprocessing pipeline and tuned an XGBoost model via a 30-trial Optuna search, reaching 0.844 ROC-AUC and 0.671 PR-AUC on a held-out test set of 1,057 customers.
- Deployed an interactive Streamlit dashboard with SHAP explainability and a profit-threshold analysis that weighs outreach cost against customer lifetime value to recommend an action cutoff.

Real-Time Facial Emotion Detection Engine

2026

- Developed a real-time webcam pipeline using OpenCV face detection and a pretrained mini-XCEPTION model (FER-2013, ~66% accuracy) to classify emotions live.
- Rendered live bounding boxes, emotion labels, and a probability bar chart across all 7 emotion categories with low-latency OpenCV rendering.

HONORS

Jack Kent Cooke Undergraduate Scholarship

National Semifinalist

National

2026

- Named a National Semifinalist for the Jack Kent Cooke Undergraduate Scholarship, recognizing top-tier high-achieving undergraduates nationwide.

SKILLS

Languages: Python, SQL, Java, TypeScript, Bash/Linux

Frameworks: PyTorch, TensorFlow, Keras, Scikit-Learn, XGBoost, LangGraph, Hugging Face Transformers

Libraries: Pandas, NumPy, OpenCV, ChromaDB, Optuna, SHAP, Ragas

Tools: AWS, GCP (Vertex AI), Docker, Kubernetes (GKE), PostgreSQL, Git/GitHub, Streamlit, Claude Code

Concepts: Generative AI, LLMs, Multimodal RAG / CRAG, PEFT/LoRA, Quantization (QLoRA), RLHF, Prompt Engineering, Computer Vision, CNNs, Agile/Scrum, OOP, MLOps